

Lem diferenciacion lebesgue l1 imp loc

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Lema 1. Sea $f \in \mathcal{L}_{loc}^1(\mathbb{R})$. Supongamos que

$$\forall g \in \mathcal{L}^1(\mathbb{R}) : \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} g(t) dt = g(x) \text{ c.t.p. } x \in \mathbb{R}.$$

$$\text{Entonces, } \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt = f(x) \text{ c.t.p. } x \in \mathbb{R}.$$

Demostración: Definimos $\forall k \in \mathbb{N} : f_k = f \cdot \mathbb{1}_{[-k,k]}$. Como $f \in \mathcal{L}_{loc}^1(\mathbb{R})$, sabemos que $\forall k \in \mathbb{N} : f_k \in \mathcal{L}^1(\mathbb{R})$. Por tanto, por hipótesis, tenemos que

$$\forall k \in \mathbb{N} : \exists E_k \subset \mathbb{R} : \left[m(E_k^c) = 0 \wedge \forall x \in E_k : \lim_{h \rightarrow 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f_k(x) dx = f_k(x) \right].$$

Tomamos $E = \bigcap_{k=1}^{\infty} E_k$, entonces $m(E^c) = m\left(\bigcup_{k=1}^{\infty} E_k^c\right) \leq \sum_{k=1}^{\infty} m(E_k^c) = 0$.

Sea $x \in E$, entonces $\exists k_0 \in \mathbb{N}$ tal que $x \in [-k_0, k_0]$. Entonces, si tomamos $h_0 = k_0 - |x| > 0$,

$$\forall h \in (0, h_0) : [x-h, x+h] \subset (-k_0, k_0) \implies \forall t \in [x-h, x+h] : \mathbb{1}_{[-k_0, k_0]}(t) = 1.$$

Por tanto, para todo $h \in (0, h_0)$, tenemos que $\forall t \in [x-h, x+h] : f(t) = f_{k_0}(t)$, luego

$$\frac{1}{2h} \int_{x-h}^{x+h} f(t) dt = \frac{1}{2h} \int_{x-h}^{x+h} f_{k_0}(t) dt.$$

Tomando límite cuando $h \rightarrow 0^+$, obtenemos

$$\lim_{h \rightarrow 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt = \lim_{h \rightarrow 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f_{k_0}(t) dt = f_{k_0}(x) = f(x). \quad \blacksquare$$

Referenciado en

- teo-diferenciacion-lebesgue

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